

Assignment – 1

Computer Networks

Subject: Computer Networks & Communication
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Q1 Connection-Oriented Network Services

A **connection-oriented network service** is a communication model that establishes a reliable, logical end-to-end link between sender and receiver *before* any data is transmitted. It is modelled after the telephone system — a caller must dial and establish a connection before speaking. Unlike connectionless services, it creates a relationship between packets, ensuring they follow the same path in sequence.

Three Phases of Communication

Phase	Name	Description
1	Connection Setup (Establishment)	Source establishes connection via handshake (e.g., TCP's three-way handshake). Parties exchange control info, synchronise sequence numbers, and initialise buffers.
2	Data Transfer	Stream of data packets sent and received in established order. Both parties can exchange data simultaneously (full-duplex).
3	Connection Termination (Release)	Connection is released after exchange to free variables, buffers, and network resources.

Key Characteristics

- **Reliability** – Uses acknowledgements and retransmissions to ensure data arrives without corruption, gaps, or duplication.
- **Orderly Delivery** – Packets arrive at the destination in the exact sequence they were sent.
- **Resource Allocation** – In virtual-circuit networks, specific resources (e.g., bandwidth) are reserved during setup.
- **Flow & Error Control** – Regulates transmission rate to prevent overwhelming the receiver and detects/corrects errors.

Examples of Connection-Oriented Protocols

Layer	Protocol	Notes
Transport	TCP, SCTP	TCP is the primary CO protocol on the Internet; SCTP handles multimedia.
Network	ATM, Frame Relay	Virtual-circuit approaches that establish a path before data is sent.
Data Link	Acknowledged CO services	Guarantee each frame is received correctly over unreliable links (e.g., satellite).

Q2 Difference Between Web Server and Proxy Server

A **web server** is the original source of content, while a **proxy server** (web cache) acts as an intermediary that retrieves and stores content on behalf of clients.

Comparison Table

Feature	Web Server	Proxy Server (Web Cache)
Primary Role	Houses and distributes web objects to clients.	Acts as intermediary; satisfies HTTP requests on behalf of origin server.
Identity	Origin server storing original files permanently.	Acts as both server (to browser) and client (to origin server).
Content Origin	Created and stored on local disk/memory.	Cached (copied) from recent requests to origin servers.
Client Interaction	Directly delivers requested pages to browser.	Intercepts request, checks local cache before contacting web server.
IP Visibility	Sees IP of requesting device.	Can hide client's IP from website, providing anonymity.

Roles in Detail

Web Server

- Implements the server side of HTTP protocol and houses web objects, each addressable by a URL.
- Waits for incoming requests, retrieves objects from storage, and sends them back.
- Typically 'always-on' with fixed, well-known IP addresses.

Proxy Server

- Maintains its own disk storage with copies of recently requested objects.
- If object is in cache → returned immediately. Otherwise → fetched from origin, stored, then forwarded.
- Reduces response times and decreases traffic on an organisation's access link.
- Can act as an application-level gateway for content filtering and user anonymity.

Q3 Different Types of Network Topologies

Topology is the schematic description of a network's arrangement — how nodes are connected through communication lines. It can be *physical* (actual cable layout) or *logical* (how the network appears to devices).

1. Mesh Topology

Every device has a dedicated point-to-point link to every other device. For n nodes, duplex links required = $n(n-1)/2$.

Advantages:

- Highly robust — if one link fails, system stays active.
- High privacy/security on dedicated lines.
- Easy fault isolation.

Disadvantages:

- Massive cabling and I/O ports required.
- Complex and expensive installation.

Example: Connecting regional telephone offices; WAN implementations.

2. Star Topology

Each node connects to a central controller (hub/switch). Devices communicate through the centre.

Advantages:

- Less expensive and easier to install than mesh.
- Failure of one link only affects that node.

Disadvantages:

- Central hub is a single point of failure.

Example: Modern LANs in offices.

3. Bus Topology

All devices linked to a single continuous cable (backbone) using drop lines and taps. Data transmitted in one direction.

Advantages:

- Most cost-effective; least cabling required.
- Easy to expand.

Disadvantages:

- Backbone fault disables entire network.
- Difficult to troubleshoot; slower.

Example: Early Ethernet networks.

4. Ring Topology

Each device has point-to-point connection with two adjacent devices forming a closed loop. Signal passes in one direction.

Advantages:

- Easy to install and reconfigure.
- Fault isolation simplified via signal monitoring.

Disadvantages:

- Failure of single computer or break disrupts entire network.

Example: Token Ring networks; MANs using SONET.

5. Tree (Star-Bus) Topology

Root node with parent-child hierarchy. Integrates multiple star topologies onto a single bus backbone.

Advantages:

- Easily expandable; managed in segments.

Disadvantages:

- More cabling than bus or star alone.

Example: Large-scale networks; broadband transmissions.

6. Hybrid Topology

Combination of two or more topologies to meet specific needs (e.g., star backbone with multiple bus networks).

Advantages:

- Inherits advantages of included topologies.
- Highly scalable and flexible.

Disadvantages:

- Complex design; often most costly to implement.

Example: Enterprise networks combining different segments.

Q4 Different Network Types

Networks are classified by geographic scale, transmission technology, and ownership.

1. PAN – Personal Area Network

Formed around an individual within ~10 m. Used to interconnect personal devices.

- Wireless PAN (WPAN): Bluetooth, WiFi, Zigbee
- Wired PAN: USB

Example: Connecting wireless headset or cordless mouse to a laptop.

2. LAN – Local Area Network

Privately owned; covers single office/building/home (~up to 2 km). High speed (100 Mbps–1 Gbps), low error rates. Uses Ethernet (wired) or WiFi (wireless).

- Resource sharing: printers, files, software

Example: Office network with shared printer; home WiFi network.

3. CAN – Campus Area Network

Multiple LANs within a specific area (university campus, military base, corporate park). 1–5 km range. Owned by a single entity.

Example: University network linking department buildings and libraries.

4. MAN – Metropolitan Area Network

Covers a city or town (up to 50 km). Larger than LAN, smaller than WAN.

- Uses FDDI, ATM, or cable technology

Example: City-wide cable TV / internet network.

5. WAN – Wide Area Network

Long-distance transmission across countries, continents, or globally (>50 km). Uses satellites, fibre-optic cables, and public telephone networks via routers.

- Switched WAN: end systems connected through routers
- Point-to-Point WAN: leased line from provider to ISP

Example: The Internet — the largest WAN in the world.

Comparison Summary

Parameter	PAN	LAN	CAN	MAN	WAN
Range	1–10 m	≤ 2 km	1–5 km	5–50 km	> 50 km
Speed	Very High	Very High	High	Average	Low
Technology	Bluetooth, USB	Ethernet, WiFi	Ethernet	FDDI, ATM	Leased line, Satellite

Parameter	PAN	LAN	CAN	MAN	WAN
Ownership	Personal	Private	Single org.	Single/Shared	Public/ISP

Q5 Short Notes: Backbone Network

A **backbone network** (core network) is a high-capacity infrastructure serving as the primary pathway for data transmission between different network segments and major locations. It forms the central part of a telecommunications or computer network, interconnecting various subnetworks.

Key Characteristics

- **High Capacity** – Handles aggregated data traffic from multiple sources.
- **Interconnection** – No individual station connects directly; stations are in LANs and the backbone connects those LANs.
- **Protocol** – Typically uses high-speed protocols such as Ethernet.
- **Medium** – Fibre optics prevalent due to high bandwidth and low attenuation.

Common Architectures

Architecture	Description	Use Case
Bus Backbone	Uses bus topology for the backbone itself.	Distribution backbone connecting different buildings.
Star Backbone (Switched)	Central switch connects all LANs; performs backbone function simultaneously.	Campus or enterprise inter-LAN connectivity.
Remote LAN Interconnection	Remote bridges link LANs via point-to-point leased lines. The P2P link acts as a LAN without stations.	Connecting geographically distant LANs.

Internet Backbone

- The Internet is a 'network of networks'; top level consists of Tier-1 ISPs.
- Tier-1 ISPs operate large international backbone networks with thousands of routers connected by high-bandwidth fibre-optic links.
- An ISP's backbone consists of long-distance transmission lines interconnecting routers at Points of Presence (POPs) in different cities.

Q6 Define Protocol. Why Do We Need Standards?

Definition of Protocol

A **protocol** is a set of rules that govern data communications and represents an agreement between communicating devices. Without a protocol, two devices may be physically connected but unable to understand each other — similar to two people speaking different languages. A protocol defines *what* is communicated, *how* it is communicated, and *when* it is communicated.

Three Key Elements of a Protocol

Element	Description
Syntax	Structure or format of the data — the order in which bits are presented. E.g., first 8 bits = sender's address, next 8 bits = receiver's address.
Semantics	Meaning of each section of bits — how a pattern is interpreted and what action to take. E.g., does an address identify a final destination or a specific route?
Timing	When data should be sent and how fast. Prevents receiver overload if sender transmits faster than receiver can process.

Why Do We Need Standards?

Standards ensure **interoperability** — products from different manufacturers work together seamlessly.

- **Interconnectivity** – Guidelines for manufacturers/vendors ensure interconnectivity in today's global marketplace.
- **Open & Competitive Markets** – Prevent a single manufacturer from monopolising technology; companies compete on quality.
- **Cost Reduction** – Agreement on standards leads to mass production, economies of scale, and lower prices for consumers.
- **Avoiding Chaos** – Without standards, there would be complete chaos — users unable to perform basic networking tasks across different systems.

Categories of Standards

Category	Definition	Examples
De Facto	Not approved by official body; adopted through widespread use.	HTTP, Bluetooth (originally)
De Jure	Legislated by an officially recognised body.	ISO, IEEE, ITU-T standards

Q7 Protocol & OSI Model Layers

The **OSI (Open Systems Interconnection)** model is a seven-layer conceptual framework proposed by ISO to facilitate communication between different systems. Each layer performs a well-defined function and provides services to the layer above it.

Layer 7	Application	Highest layer; closest to user.
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- Provides interfaces/support for network services.
- Allows users to access network resources via file transfer (FTAM), email, directory services.
- Identifies communication partners and determines resource availability.

Protocols/Devices: HTTP, FTP, SMTP, DNS

Layer 6	Presentation	Concerned with syntax and semantics of information exchanged.
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- Translation (encoding conversion) between different data representations.
- Encryption / Decryption for security.
- Data compression to reduce bits transmitted.

Protocols/Devices: SSL/TLS, JPEG, MPEG

Layer 5	Session	Network dialog controller.
• Establishes, maintains, and synchronises interactions between systems. • Dialog control: half-duplex or full-duplex communication. • Synchronisation: adds checkpoints to data streams for crash recovery. Protocols/Devices: NetBIOS, RPC		
Layer 4	Transport	Known as the 'Heart of OSI'. Responsible for process-to-process delivery.
• Ensures messages arrive intact, in order, with no duplication. • Service-point (port) addressing to identify specific processes. • Segmentation and reassembly; end-to-end flow and error control. Protocols/Devices: TCP (connection-oriented), UDP (connectionless)		
Layer 3	Network	Responsible for source-to-destination delivery across multiple networks.
• Logical addressing using IP addresses. • Routing: determines best path based on network conditions. • Packetising and congestion control. Protocols/Devices: IP, ICMP, OSPF, BGP — Device: Router		
Layer 2	Data Link	Responsible for node-to-node (hop-to-hop) delivery; makes physical link appear error-free.
• Divides bit stream into frames. • Physical addressing (MAC addresses). • Flow control and error control for damaged frames. • Sublayers: LLC (Logical Link Control) and MAC (Media Access Control). Protocols/Devices: Ethernet, Wi-Fi (802.11), PPP — Device: Switch, Bridge		
Layer 1	Physical	Lowest layer; moves individual bits from node to node over physical medium.
• Mechanical/electrical specs: voltages, data rates, connectors. • Bit synchronisation, bit rate control, physical topologies, transmission modes. Protocols/Devices: Cables, Hubs, Repeaters, Modems		

Q8 TCP/IP Model & Comparison with OSI Model

The **TCP/IP model** (Internet protocol suite) is a practical networking framework developed for ARPANET/Internet. While originally four layers, modern interpretations use five layers.

TCP/IP Layers (5-Layer Model)

Layer	Responsibilities	Key Protocols
5 – Application	Provides network services to end-users and applications. Combines OSI Application, Presentation, and Session functions.	HTTP, SMTP, FTP, DNS

Layer	Responsibilities	Key Protocols
4 – Transport	Process-to-process delivery. TCP: reliable, connection-oriented, flow/congestion control. UDP: connectionless, no reliability, suitable for speed-sensitive tasks.	TCP, UDP
3 – Network (Internet)	Logical addressing and routing across different networks. Determines best path from source to destination.	IP (primary), ICMP, ARP
2 – Data Link	Node-to-node delivery; makes physical link error-free. Organises bits into frames; MAC addressing; flow and error control.	Ethernet, Wi-Fi
1 – Physical	Actual transmission of raw bits over physical media. Defines hardware specs, data rates, physical topologies.	Cables, Hubs, Repeaters

OSI vs TCP/IP Comparison

Feature	OSI Model	TCP/IP Model
Layers	7 layers	4 or 5 layers
Nature	Theoretical / reference model	Practical / implementation model
Development	Proposed by ISO in 1984	Developed by DoD for ARPANET
Upper Layers	Separate Session, Presentation, Application	Combined into single Application layer
Lower Layers	Distinct Physical and Data Link	Originally combined into Link layer
Reliability	Guaranteed delivery at theoretical level	Best-effort at IP layer (not guaranteed)
Network Layer Service	Both connection-oriented and connectionless	Connectionless only

Key Mapping: TCP/IP Application layer ≈ OSI Session + Presentation + Application. TCP/IP Transport layer handles some OSI session management tasks.

Q9 Two Network Architectures in Practice

In practice, two predominant architectural paradigms are used in modern network applications.

1. Client-Server Architecture

A central, always-on host called the **server** provides services to many **clients**. The server acts as a central controller managing security and access to data.

- Clients do NOT communicate directly with each other.
- Server has a fixed, well-known IP address so clients can always contact it.
- Centralised data management makes backups easier.
- Suitable for small to very large-scale networks using powerful data centres.

Examples: World Wide Web (HTTP), Email (SMTP), FTP.

2. Peer-to-Peer (P2P) Architecture

Minimal or no reliance on dedicated servers. Application exploits direct communication between pairs of intermittently connected hosts called **peers**. Every computer has equal privileges.

- Every computer acts as both file server and client simultaneously.
- Self-scalability: new peers add both demand and service capacity.
- Highly cost-effective — no expensive server infrastructure needed.
- Best for small environments (≤ 10 computers) or highly decentralised global apps.

Examples: BitTorrent (file sharing), Skype (Internet telephony/video).

3. Hybrid Architecture

Some applications combine both models. E.g., instant messaging uses a central server to track IP addresses (Client-Server), but messages are sent directly between users (P2P).

Q10 Classify Networks on the Basis of Size

Computer networks are primarily classified based on their geographical scale or physical size.

Network	Range	Speed	Technology	Example
PAN	1–10 m	Very High	Bluetooth, USB	Wireless headset to laptop
LAN	≤ 2 km	Very High (100 Mbps–1 Gbps)	Ethernet, WiFi	Office/home network
CAN	1–5 km	High	Ethernet	University campus network
MAN	5–50 km	Average	FDDI, ATM, Cable	City-wide cable/internet
WAN	> 50 km	Low	Leased line, Satellite	The Internet

Other Specialised Types

- **VPN (Virtual Private Network)** – Secure private network using the public internet to connect remote sites through virtual connections.
- **Internetwork** – Formed when two or more distinct networks (LAN + WAN) are connected using routers.

Q11 Client/Server Network vs Peer-to-Peer Network

Client-Server Network

Clear separation between the central **server** (always-on, provides resources) and **clients** (end-user devices that request resources). Clients do not communicate directly with each other.

- **Server:** Handles security, data management, and resource allocation. Fixed, well-known IP address.
- **Clients:** PCs, smartphones, laptops that initiate requests.
- **Examples:** WWW (HTTP), Email (SMTP).

Peer-to-Peer Network

No dedicated servers. Peers communicate directly with each other, acting as both client and server simultaneously.

- **Equal Roles:** Any node can request or provide data.
- **Self-Scalability:** More peers = more capacity.
- **Examples:** BitTorrent, Skype.

Key Differences

Feature	Client-Server	Peer-to-Peer (P2P)
Role Differentiation	Clear server/client separation	No fixed roles; peers act as both
Communication	Clients ↔ Server only	Peers communicate directly
Data Management	Centralised on server	Decentralised; each peer has own data
Scalability	May overload without data centres	Self-scaling as users join
Cost	Expensive (dedicated hardware, admins)	Inexpensive (no central infrastructure)
Reliability	Server = single point of failure	Highly robust; peer failure = minimal impact
Environment	Small to very large-scale networks	Small networks or decentralised global apps

Q12 Explain LAN with Example. How Is It Different from PAN?

Local Area Network (LAN)

A **LAN** is a privately owned network connecting computers and peripheral devices within a restricted geographical area — typically a single office, building, or campus.

- **Purpose** – Resource sharing: hardware (printers), software, and data among multiple users.
- **Speed** – 10 Mbps to 100 Mbps; modern versions reach 1 Gbps or 10 Gbps.
- **Technology** – Wired: Ethernet (IEEE 802.3). Wireless: WiFi (IEEE 802.11).
- **Examples** – Office network with workstations sharing a networked printer; home network with laptop, PC, and router.

Differences Between LAN and PAN

Feature	PAN	LAN
Scale / Range	1–10 metres around an individual	Single building/campus; up to ~2 km
Primary Use	Interconnect personal devices of one person	Share resources among multiple users in a group
Technology	Bluetooth, Zigbee, USB	Ethernet cables, WiFi
Examples	Wireless mouse/keyboard to laptop	School computer lab; entire office building
Maintenance	Very easy; simple device pairing	Easy–Moderate; needs admin for larger setups
Connectivity	Personal space communication only	Gateway to broader Internet for users

In summary: A PAN is designed for the convenience of a single person and their gadgets, while a LAN provides robust, high-speed communication and shared resources for a group of people in a fixed location.